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THE MAGNETIZATION OF PRE-CAMBRIAN ROCKS FROM THE TÄRENDÖ DISTRICT, NORTHERN SWEDEN

By J. D. CORNWELL

ABSTRACT. — The Pre-Cambrian rocks of Northern Sweden generally have a weak remanent magnetization induced by the present geomagnetic field. Exceptions to this are the gabbro and adjacent country rocks near the village of Tärendö, Norrbotten, for these retain stable, well grouped directions of magnetization significantly different from that of the present field. These are thought to represent the direction of the palaeomagnetic field 1800—2000 m. y. ago — the oldest result so far obtained in Europe. The palaeomagnetic pole position calculated from the directions of the samples after magnetic cleaning lies at 45°N, 132°W. Although this isolated result needs to be followed up by more investigations it is significant to note that an almost identical result has previously been obtained from the Ylivieska gabbro in Finland.

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A systematic investigation of the magnetic properties of the Pre-Cambrian rocks of Norrbotten county, Northern Sweden, has been carried out to assist in the interpretation of aeromagnetic and ground magnetic surveys. The majority of the rocks have a weak remanent magnetization coincident with the present geomagnetic field, suggesting that the original thermo-remanent magnetization (TRM) has decayed to be replaced in part by a viscous remanent magnetization (VRM) induced by the present field. Some samples collected from one exposure of the gabbro near the village of Tärendö by the Kalixälv (Fig. 1) proved to be exceptional in that their remanent magnetization was relatively strong and did not coincide with the geomagnetic field. Similar features were subsequently found for other sites on the gabbro and on the adjacent migmatite and syenite.

The closer investigation of the magnetization of the Tärendö gabbro was undertaken with a view also of explaining the high magnetic anomalies recorded in this area during the aeromagnetic survey. The map based on data recorded at a height of 30 m above the surface shows clearly the shape of the gabbro sub-outcrop and a large number of fault lines, allowing the construction of a far more accurate geological map than would have been possible from the limited number of outcrops available in the area (Fig. 2). Interpretation of north-west to south-east profiles across the main part of the gabbro body suggests that it has a saucer-like form extending down to a depth of 2 km. This form gives rise to a negative magnetic anomaly of 4000 gammas along the north-western margin of the intrusion (Cornwell 1964) while positive anomalies of up to 8000 gammas occur over the gabbro itself.

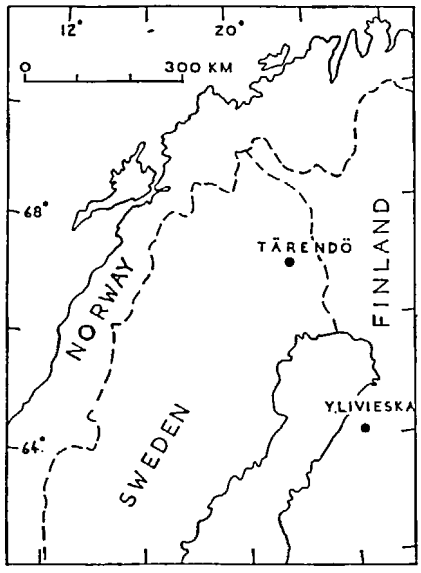


Fig. 1. Locations of the Tärendö and Ylivieska gabbros.

The Tärendö gabbro was described briefly by Eriksson (1954) as being a titano-magnetite-bearing type of Karelian age. Like most of the Pre-Cambrian rocks in Northern Sweden it has not been dated radiometrically but on the magnetic map the gabbro appears to have been invaded by the 'younger granite' of the late kinematic granite series lying to the north-west and at one of the sites visited (T87) it has been intersected by granite dykes. The minimum age of the gabbro is therefore that of the late kinematic granites — 1775 to 1845

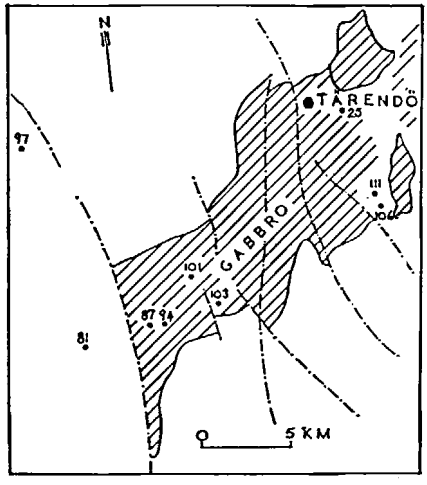


Fig. 2. Simplified map of the Tärendö gabbro and locations of the collecting sites T25 —T111.

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m. y. according to Welin and Blomquist (1966) — and the maximum age is believed to be 2000 m. y. (Welin, private communication).

In addition to the five sites on the gabbro, samples were collected from the younger granites to the north of the intrusion (Table 1; Fig. 2), the migmatite granites to the east and west, and from the granite dykes intruded into the gabbro at site T 87. All the sites are fresh roadside exposures. The remanence and susceptibility of the orientated hand samples were measured with a four probe 'Oerstedmeter' soon after collection and these results are listed in Table 1.

The mean directions of magnetization (D°, I°) have been computed for each site from the data for n samples and are relauve to the original positions of the samples in the field. k and α° are measures of the scatter of the directions according to the statistical method of Fisher (1953) in which each sample vector is given unit length. The susceptibilities of the rocks listed in Table 1 are typical for the respective types but the remanence for 7 of the 9 sites is abnormally high, resulting in Q-values (the ratio of the remanence to the magnetization induced in a field of 0.5 oersted) between 1 and 4. Typical Q-values for the Pre-Cambrian rocks of Norrbotten lie between 0.1 and 0.4. This feature, together with the tight grouping of the directions of magnetization about means significantly different from that of the present field suggest that the magnetization is stable and could be the original TRM of these igneous and metamorphic rocks. The fact that the rocks adjacent to the gabbro in the east and even the dykes which intrude it have similar directions of magnetization and high Q-values is of particular interest. Two of the sites in Table 1, T 81 and T 97, yielded rocks with the low Q-values and directions of magnetization typical for most of the Pre-Cambrian rocks of Northern Sweden.

Further evidence for the stability of the NRM of the gabbro samples is provided by their resistance to demagnetization by ac fields. This method of magnetic cleaning is commonly applied to rocks used to determine palaeomagnetic field directions and is particularly useful for removing the VRM component acquired in recent times. Because of its nature VRM can be destroyed by ac magnetic fields of 100 to 150 oersteds, leaving behind the harder components of the original magnetization. Samples from the gabbro showed only small changes in directions of magnetization after demagnetization in ac fields of 150 oersteds and in some cases the scatter even increased (Table 1), probably due to experimental errors. The proportion of the remanence destroyed in ac fields of up to 150 oersteds (the limit of the apparatus available) is small, particularly when compared with the rapid intensity decrease (Fig. 3) shown by another Norrbotten gabbro which had an original direction coincident with the geomagnetic field and a low Q-value. The VRM component is large in this second sample and more typical of the Pre-Cambrian rocks of the area

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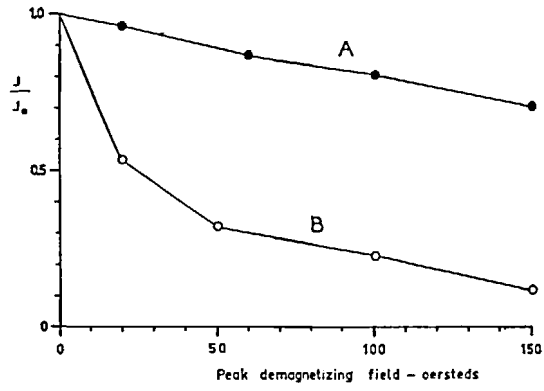


Fig. 3. Normalized ac demagnetization curves for (A) a sample from site T87 on the Tärendö gabbro and (B) a sample from the Vittangi gabbro.

but in the Tärendö gabbro and probably in the rocks from 3 other sites listed in Table 1 the VRM is weak. The mean directions of magnetization for the 5 sites on the gabbro after magnetic cleaning have been averaged to give a total mean for the intrusion of 340°+41° (Table 2).

The directions of the migmatite and syenite adjacent to the gabbro and the acid dykes at site T 87 are similar to these of the gabbro and the 12° difference in their means (Table 2) is statistically not significant. This similarity suggests that they were magnetized simultaneously, but there is no such evidence for the rocks at sites T 81 and T 97, for these have magnetic properties suggesting the predominance of the VRM component. It is suggested that these rocks were formed at a different, possible later, time to the gabbro and adjacent rocks to the east and their position west of the large fault bounding the western margin of the gabbro is therefore significant (Fig. 2).

It seems most likely that the magnetization of the gabbro and migmatization dates from the time of the original formation of these rocks and was largely unaffected by the subsequent formation of the late kinematic granites and syenites. In favour of this view are the absence of similar directions of magnetization in the younger granites and syenites at site T 97 and elsewhere in Norrbotten and the necessity of postulating that the entire gabbro intrusion had been heated to a high enough temperature to allow remagnetization. Because of their similar directions of magnetization the acid dykes in the gabbro (T 89) should be connected genetically with the syenite collected at site T 106. This does not necessarily exclude the syenite and the gabbro from being members of the younger granite series as the late kinematic rocks were probably formed at different times over an extended period of time. Ödman (1957) in fact suggests that some of the gabbros in Norrbotten are members of the series.

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TABLE 1. *Magnetic properties and directions of magnetization of the gabbro and acid rocks of the Tärendö district.*

Site No.	Rock type	RAK co-ordinates		Susceptibility 10^{-3} gauss/oersted	Remanence 10^{-3} gauss	NRM_0 Q	NRM_0 D°	NRM_0 I°	NRM_0 k	NRM_0 α°_{95}	NRM 150 oersteds D°	NRM 150 oersteds I°	NRM 150 oersteds k	NRM 150 oersteds α°_{95}	n
T 25	Gabbro	17964	74683	6.60	3.89	1.18	337	+41	126	11	331	+28	54	17	3
T 87	Gabbro	17859	74562	5.79	8.20	2.83	318	+52	65	8	330	+44	32	12	6
T 94	Gabbro	17868	74593	8.43	14.22	3.37	357	+38	300	7	347	+40	9990	1	3
T 101	Gabbro	17880	74586	11.87	10.43	1.76	334	+46	—	—	340	+38	—	—	2
T 103	Gabbro	17898	74574	4.88	10.61	4.35	348	+45	338	7	355	+49	310	7	3
T 81	Migmatite	17824	74549	4.48	0.65	0.29	57	+50	12	24	—	—	—	—	5
T 89	Granite dykes	as T 87		0.66	0.53	1.59	333	+51	90	10	—	—	—	—	4
T 97	Granite	17787	74659	1.29	0.19	0.30	121	+87	15	33	—	—	—	—	3
T 106	Syenite	17986	74629	3.69	2.91	1.58	323	+31	118	9	—	—	—	—	4
T 111	Migmatite	17978	74636	5.53	3.64	1.31	320	+45	6	41	—	—	—	—	4

TABLE 2. *Mean directions of magnetization and palaeomagnetic pole positions for the Tärendö district, Sweden, and the Ylivieska gabbro, Finland.*

Site group	Latitude	Longitude	No. of Sites	Directions of magnetization D°	I°		k	α°_{95}	Palaeomagnetic pole position Latitude	Longitude
Tärendö gabbro	67.1°N	22.5°E	5	340	+41	*	30	14	45°N	132°W
Tärendö acid rocks	67.1°N	22.5°E	3	324	+42		51	18	42°N	112°W
Ylivieska gabbro	64.1°N	24.5°E	1	341	+41		145	4	47°N	129°W

* — after ac demagnetization

The age of the gabbro and its magnetization therefore lies between that of the late kinematic granites (about 1800 m. y.) and 2000 m. y. Systematic demagnetization experiments on other rocks from Norrbotten might reveal directions of magnetization similar to that of the Tärendö gabbro but in none of the other groups of samples so far examined has such a strong remanence been preserved. This is difficult to explain unless it is assumed that the gabbro was formed at an interval in time when the geomagnetic field was particularly strong. An alternative explanation that the gabbro is considerably younger than the other rocks in Norrbotten and therefore more likely to retain a strong remanence is not supported by geological evidence.

Palaeomagnetic data from rocks with a known age as great as that of the Tärendö gabbro are not available for Europe but it is of interest to note that Puranen (1960), in discussing the general magnetic properties of rocks, gives the example of a gabbro at Ylivieska, Finland (Fig. 1) which has a remanent magnetization similar to that at Tärendö. The virtual palaeomagnetic pole position calculated from the mean of the 13 directions measured by Puranen agrees with that for the Tärendö gabbro (Table 2), suggesting that the two intrusions were formed contemporaneously, but it must be pointed out that the effects of ancient secular variation have probably not been averaged out in Puranen's data.

Other results from the Pre-Cambrian of Europe are few (Irving 1964) and the Lower Torridonian sandstones of western Scotland are about 1000 m. y. younger than the Tärendö gabbro although they indicate a similar palaeomagnetic pole position of 35°N, 118°W. The nearest estimate in time of a palaeomagnetic pole position comes from a suite of diabase dikes in Hame, central Finland, with an age of between 1640 and 1800 m. y. (Neuvonen 1967). These yield a pole position of 67°N, 72°E, far removed from that of the Tärendö gabbro and also from the younger dikes of Satakunda and the Vaasa archipelago, Finland (Neuvonen 1965, 1966), although Neuvonen points out that the directions of the Hame dikes lie close to that of the present field and could possibly be due to VRM. The Jotnian dikes examined by Neuvonen (1965, 1966) are some 500 m. y. younger than the Tärendö gabbro and yield pole positions just north of the Equator about the 160°E line of longitude. It is therefore obvious that more data is needed before the polar wandering curve for Europe can be extended back into the early Pre-Cambrian but it must be characterized by large movements and to some extent repeat the positions discovered earlier (Irving 1964) from post- and late Pre-Cambrian rocks.

In Africa, investigations on the Modipe gabbro have shown that the geomagnetic field existed already 2600 m. y. ago (Evans & McElhinney 1966) and it has been possible to construct a preliminary polar wandering curve for

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the Pre-Cambrian of that continent (Evans 1967). Evans and McElhinney explained the preservation of a very stable magnetization in the Modipe gabbro as the result of the occurrence of elongated single domains of magnetite with a high coercivity due to their shape anisotropy. A similar explanation in terms of the mineralogy for the magnetization of the Tärendö district rocks does not seem likely as several different rock types are involved. In the Canadian shield, rocks as old as the Tärendö gabbro have been investigated and it has been possible to use palaeomagnetism as a means of correlation and relative age determination (Strangway 1964).

Apart from their palaeomagnetic interest these results from the Tärendö gabbro have some significance in the interpretation of the aeromagnetic maps over the Pre-Cambrian in Norrbotten for they show that large magnetic anomalies due to the presence of a strong remanent magnetization can occur. These can be of the same amplitude as those recorded over small deposits of magnetite in which the magnetization is largely that induced by the geomagnetic field. Such cases are fortunately rare.

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